

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 05 · Constant Acceleration in 2D

Questions

Student book · questions with writing space

- 11 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 05

Constant Acceleration in 2D

Your classified route

4

QUESTIONS

38

MARKS

1

FAMILIES

3

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 05 | FAMILY 01

Constant-acceleration vector kinematics

WHAT THIS FAMILY ASKS

Use componentwise displacement, velocity and acceleration relationships for motion in a plane.

VARIANT ROUTE

V01 Recover constant acceleration

V02 Update velocity with $v = u + at$

V03 Two accelerated particles with equal speed/direction conditions

4 classified questions

38 total marks

6. A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

Find the speed of P at time $t = 3.5$ seconds.

(6)

DR ESAM AHMED

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

(a) Find the value of T .

(6)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

A particle P is moving with constant acceleration. At 2pm, the velocity of P is $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$ and at 2.30pm the velocity of P is $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time T hours after 2pm, P is moving in the direction of the vector $(-\mathbf{i} + 2\mathbf{j})$

Another particle, Q , has velocity $\mathbf{v}_Q \text{ km h}^{-1}$ at time t hours after 2pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and μ is a constant.

Given that there is an instant when the velocity of P is equal to the velocity of Q ,

(b) find the value of μ .

(3)

DR ESAM AHMED

4. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ms}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ms}^{-1}$

- (a) Find the velocity of P when $t = 5$ s, giving your answer in terms of $\mathbf{i}$, $\mathbf{j}$ and λ .

(2)

The speed of P when $t = 5$ s is 13ms^{-1}

- (b) Show that

$$25\lambda^2 - 42\lambda - 16 = 0$$

(3)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively.]

A particle P moves with constant acceleration $(-\lambda\mathbf{i} + 2\lambda\mathbf{j})\text{ms}^{-2}$, where λ is a positive constant.

At time $t = 0$, the velocity of P is $(5\mathbf{i} - 8\mathbf{j})\text{ms}^{-1}$

- (c) Find the direction of motion of P when $t = 4\text{ s}$, giving your answer as a bearing to the nearest degree.

(5)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

Two boats, A and B , are each moving with constant acceleration.

At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

(a) Find the magnitude of the acceleration of A .

(2)

DR ESAM AHMED

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

(b) Find the acceleration of B , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)

(c) Find the velocity of B at time $t = 0$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (2)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

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At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

At time T_1 hours after noon, both boats are moving with the same speed.

(d) Find the exact value of T_1

(4)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

Two boats, A and B , are each moving with constant acceleration.

At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

At time T_2 hours after noon, both boats are moving in the same direction.

(e) Show that $3T_2^2 + pT_2 + q = 0$, where p and q are integers to be found.

(3)

DR ESAM AHMED